

Software Requirements Specification (SRS)

Problem Statement No. 15 — Railway Reservation System

Project: Railway Reservation System (RRS)

Version: 1.0

Authors: Syed Obaid Faraz (PES2UG24CS546) · Simon C A (PES2UG24CS503) · Sudha (PES2UG24CS529) · Tanisha Khorla (PES2UG24CS551) — Group 5

Date: 04-09-2026

Status: Draft — Submitted for Review

Revision history

Version	Date	Author	Change summary	Approval
1.0	04-09-2026	Group 5	Initial SRS submission for Problem Statement No. 15: Railway Reservation System (core-feature scope)	Pending

Approvals

Role	Name	Signature / Email	Date
Course Coordinator			

Table of Contents

1. Introduction
2. Overall description
3. External interfaces
4. System features (detailed)
5. Non-functional requirements (detailed)
6. Quality attributes & Acceptance tests
7. System models and diagrams (UML Use-Case Diagrams)
8. Requirements Traceability Matrix (RTM)

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional and non-functional requirements for a Railway Reservation System (RRS) — a simplified online platform that lets passengers search trains, book and cancel tickets, and check PNR status, and lets an admin maintain the list of available trains. The scope is intentionally kept to a small, clearly implementable core, since project evaluation for this course focuses on engineering workflow (Git branching/commits, Jira tracking) rather than feature breadth (Problem Statement No. 15, Group 5).

1.2 Scope

The RRS covers: passenger registration and login, searching trains by source/destination/date, viewing seat availability and fare, booking a ticket for up to 4 passengers with a single payment step, cancelling a booking, checking PNR status, viewing booking history, and basic admin management of trains (add/update/delete, view bookings). It excludes multi-quota fare logic, Tatkal booking, automatic waitlist/RAC promotion, offline TTE verification, and large-scale concurrency handling — these are deliberately out of scope to keep the implementation focused and demoable within the project timeline.

1.3 Audience

Developers, QA reviewers, project evaluators/course coordinators, and passengers (as end users of the described features).

1.4 Definitions

List of acronyms: PNR (Passenger Name Record), UI (User Interface), API (Application Programming Interface), JWT (JSON Web Token), HTTPS (Hypertext Transfer Protocol Secure), SRS (Software Requirements Specification).

2. Overall description

2.1 Product perspective

The RRS is a web-accessible self-service application used directly by passengers, with a separate admin view for train management. It comprises a browser-based front end, a backend application server implementing booking rules, and a relational database storing users, trains and bookings. A single payment gateway (or a simulated payment step) is used to complete a booking.

2.2 Major product functions

- User registration and login
- Train search and seat/fare display
- Ticket booking with payment
- Ticket cancellation
- PNR status enquiry and booking history
- Admin management of trains and their bookings

2.3 User roles and characteristics

- Passenger: general user booking train tickets; expects a simple, guided booking flow.
- Admin: maintains the list of trains available for booking and reviews bookings made against them.

2.4 Operating environment

Web application deployed on a standard hosting environment (e.g., a cloud VM or platform-as-a-service) reachable over the internet or campus network for demo purposes; accessed via any modern desktop or mobile browser.

2.5 Constraints

Single payment gateway integration (or simulated payment for demo); single currency and language (English); project timeline and team size limit scope to the core booking flow described in Section 1.2.

3. External interface requirements

3.1 User interfaces

A responsive web application with a simple flow: search → select train/class → enter passenger details → pay → confirmation. Clear, readable forms and error messages for invalid input.

3.2 Hardware interfaces

- None beyond a standard desktop or mobile device with a web browser and internet connectivity.

3.3 Software interfaces

- Payment Gateway API (or a simulated payment endpoint) for completing bookings
- Application database (stores users, trains, bookings)

3.4 Communications

- HTTPS for all communication between browser and server
- Standard request/response REST API calls between frontend and backend

This document specifies 19 functional requirements (Section 4), 5 non-functional requirements (Section 5), 2 security objectives and 5 security requirements (Section 5.1) — satisfying the minimum project criteria of ≥ 15 FRs, ≥ 5 NFRs, 2 security objectives and 5 security requirements, while keeping the feature set intentionally simple.

4. System features (detailed)

Each requirement below includes acceptance criteria and a reference test case. IDs follow RRS-F-###.

4.1 User Registration & Authentication

Description: Allow passengers to create an account and log in.

Req ID	Requirement (shall...)	Type	Priority	Source/ Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
RRS-F-001	The system shall allow a new user to register with name, email and password.	Functional	High	Passenger	AC-RRS-F-001: A new account is created and the user can immediately log in. Test: TC-Reg-01	—
RRS-F-002	The system shall allow a registered user to log in using email and password.	Functional	High	Passenger	AC-RRS-F-002: Correct credentials grant access to the dashboard; incorrect ones show an error. Test: TC-Auth-01	—
RRS-F-003	The system shall allow a logged-in user to log out, ending their session.	Functional	Medium	Passenger	AC-RRS-F-003: After logout, protected pages are no longer accessible without logging in again. Test: TC-Auth-02	—
RRS-F-004	The system shall allow a user to reset a forgotten password via a reset link sent to their registered email.	Functional	Medium	Passenger	AC-RRS-F-004: Clicking the emailed link lets the user set a new password and log in with it. Test: TC-Auth-03	—

4.2 Train Search & Seat Availability

Description: Let passengers find trains and check seats and fare.

Req ID	Requirement (shall...)	Type	Priority	Source/ Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
RRS-F-010	The system shall allow a user to search for trains by source station, destination station and journey date.	Functional	High	Passenger	AC-RRS-F-010: Search returns all trains running that route on that date. Test: TC-Search-01	—
RRS-F-011	The system shall display the number of available seats per class (Sleeper, AC) for each train in the results.	Functional	High	Passenger	AC-RRS-F-011: Displayed seat count matches the current database value. Test: TC-Search-02	—
RRS-F-012	The system shall display the fare for each class when a passenger selects a train.	Functional	Medium	Passenger	AC-RRS-F-012: Fare shown matches the stored fare for that train/class. Test: TC-Fare-01	—

4.3 Ticket Booking & Payment

Description: Allow passengers to book a seat and pay for it.

Req ID	Requirement (shall...)	Type	Priority	Source/ Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
RRS-F-	The system shall allow a	Functional	High	Passenger	AC-RRS-F-020: Booking form	—

Req ID	Requirement (shall...)	Type	Priority	Source/ Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
020	logged-in passenger to select a train and class, and enter details (name, age, gender) for up to 4 passengers per booking.	I			rejects submissions with 0 or more than 4 passengers. Test: TC-Book-01	
RRS-F-021	The system shall check seat availability before confirming a booking and reject the booking if not enough seats remain.	Functiona I	High	Passenger / Business	AC-RRS-F-021: Booking attempts beyond available seats are rejected with a clear message. Test: TC-Book-02	—
RRS-F-022	The system shall process payment through a single integrated payment gateway (or a simulated payment step for demo purposes) before confirming a booking.	Functiona I	High	Payment Gateway	AC-RRS-F-022: Booking is confirmed only after the payment step returns success. Test: TC-Pay-01	Single gateway integration is sufficient
RRS-F-023	The system shall generate a unique PNR and reduce the available seat count for that train/class once a booking is confirmed.	Functiona I	High	Passenger / Business	AC-RRS-F-023: A PNR is generated and seat count decreases by the number of passengers booked. Test: TC-Pay-02	—

4.4 Ticket Cancellation

Description: Let passengers cancel a booking they made.

Req ID	Requirement (shall...)	Type	Priority	Source/ Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
RRS-F-030	The system shall allow a logged-in passenger to cancel one of their own confirmed bookings.	Functiona I	High	Passenger	AC-RRS-F-030: Selected booking status changes to 'Cancelled'. Test: TC-Cancel-01	—
RRS-F-031	The system shall increase the available seat count for that train/class when a booking is cancelled.	Functiona I	High	Business	AC-RRS-F-031: Seat count increases by the number of passengers on the cancelled booking. Test: TC-Cancel-02	—
RRS-F-032	The system shall display a cancellation confirmation message to the passenger after a successful cancellation.	Functiona I	Low	Passenger	AC-RRS-F-032: Confirmation message/screen is shown immediately after cancellation. Test: TC-Cancel-03	—

4.5 PNR Status & Booking History

Description: Let passengers check the status of a booking.

Req ID	Requirement (shall...)	Type	Priority	Source/ Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
RRS-F-040	The system shall allow a user to check the status of a booking (Confirmed / Cancelled) by entering a valid PNR number.	Functiona I	High	Passenger	AC-RRS-F-040: Correct status and journey details are shown for a valid PNR; an error is shown for an invalid one. Test: TC-PNR-01	—

Req ID	Requirement (shall...)	Type	Priority	Source/ Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
RRS-F-041	The system shall allow a logged-in passenger to view a list of their own past and upcoming bookings.	Functional	Medium	Passenger	AC-RRS-F-041: Booking history list matches the passenger's bookings, most recent first. Test: TC-History-01	—

4.6 Admin — Train Management

Description: Let an admin maintain the list of trains available for booking.

Req ID	Requirement (shall...)	Type	Priority	Source/ Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
RRS-F-050	The system shall allow an authorised admin to add a new train with train number, name, source, destination, date, class and seat count.	Functional	High	Admin	AC-RRS-F-050: New train appears in passenger search results once added. Test: TC-Admin-01	—
RRS-F-051	The system shall allow an authorised admin to update or remove an existing train's details.	Functional	High	Admin	AC-RRS-F-051: Updated/removed train details are reflected in subsequent searches. Test: TC-Admin-02	—
RRS-F-052	The system shall allow an authorised admin to view the list of bookings made for a given train.	Functional	Medium	Admin	AC-RRS-F-052: Booking list shown matches confirmed bookings for that train. Test: TC-Admin-03	—

5. Non-functional requirements (detailed)

NFRs below are measurable and tied to test plans. IDs RRS-NF-####.

Req ID	Requirement	Category	Priority	Acceptance criteria / Measurement
RRS-NF-001	Train search results shall be returned within 3 seconds under normal load during a demo/classroom setting.	Performance	Medium	Manual timing check during demo. Test: TC-Perf-01
RRS-NF-002	The user interface shall be simple and consistent, allowing a first-time user to complete a booking without external help.	Usability	High	Informal usability walkthrough with a test user. Test: TC-UX-01
RRS-NF-003	The system shall not allow two bookings to be confirmed for more seats than are available on a train/class (no over-booking).	Reliability	High	Test booking attempts up to and beyond seat capacity; last attempt is correctly rejected. Test: TC-Integrity-01
RRS-NF-004	The system shall remain available and functional for the duration of project demonstrations and evaluations.	Availability	Medium	System is up and reachable during scheduled demo slots. Test: Manual check
RRS-NF-005	The codebase shall be organised into clearly separated modules (frontend, backend, database) to support independent development and review.	Maintainability	Medium	Code review confirms modular structure aligned with repo folder layout. Test: Manual review

5.1 Security

5.1.1 Security Objectives

1. Confidentiality — Protect user passwords and personal booking details from unauthorised access.
2. Integrity — Ensure booking and seat-count data cannot be tampered with or corrupted by unauthorised users.

5.1.2 Security Requirements

Req ID	Requirement (shall...)	Type	Priority	Acceptance criteria / Test case ref
RRS-SR-001	User passwords shall be stored in hashed form (e.g., bcrypt) and never stored or logged in plain text.	Security	High	Verified via security test plan / code review (see Section 6).
RRS-SR-002	The system shall use token-based session authentication (e.g., JWT) so that booking and admin actions require a valid logged-in session.	Security	High	Verified via security test plan / code review (see Section 6).
RRS-SR-003	The system shall validate and sanitise all user input on the server side to prevent SQL injection and cross-site scripting (XSS).	Security	High	Verified via security test plan / code review (see Section 6).
RRS-SR-004	Admin-only actions (add/update/delete train, view all bookings) shall be restricted to users with an Admin role.	Security	High	Verified via security test plan / code review (see Section 6).
RRS-SR-005	All communication between the browser and the server shall use HTTPS.	Security	Medium	Verified via security test plan / code review (see Section 6).

6. Quality attributes & Acceptance tests

- Exit criteria for acceptance: all high-priority functional requirements implemented and verified, no critical NFR failures, and the RTM (Section 8) shows all test cases passed.
- Acceptance test suites: Registration & Login, Search, Booking & Payment, Cancellation, PNR/History, Admin, and Security tests.

7. System models and diagrams

7.1 UML Use-Case diagrams

Two UML use-case diagrams are provided below, covering (a) the passenger booking journey and (b) admin train management.

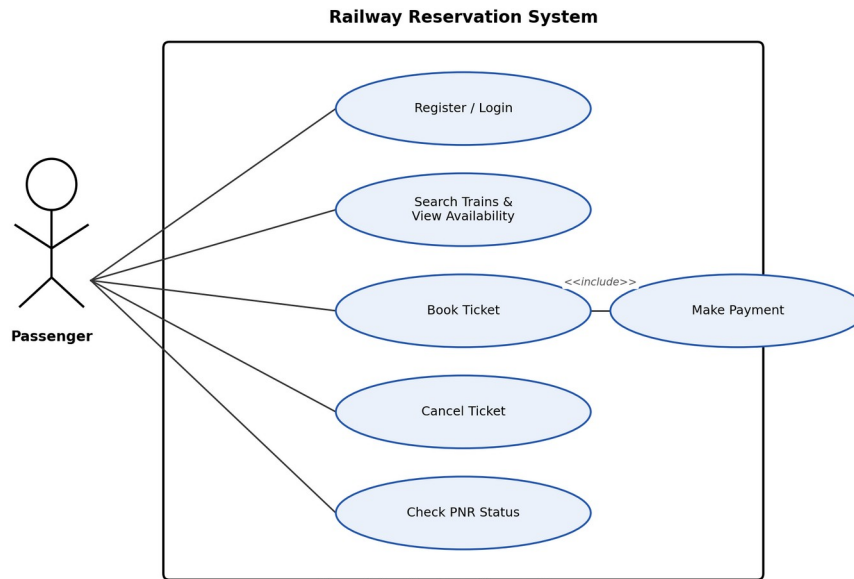


Figure 7.1 — Use-Case Diagram: Passenger

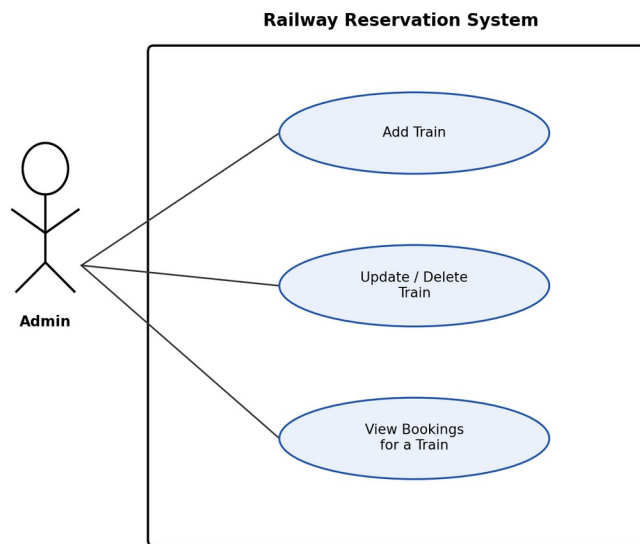


Figure 7.2 — Use-Case Diagram: Admin

8. Requirements Traceability Matrix (RTM)

Status legend: N = Not started, P = In progress, A = Passed/Accepted. All items are currently “N” pending implementation and test execution.

Req ID	Requirement (short)	Section ref	Module	Test case(s)	Status (N/P/A)	Comments
RRS-F-001	The system shall allow a new user to register with name, ...	4.1	AuthModule	Test:TC-Reg-01	N	
RRS-F-002	The system shall allow a registered user to log in using ...	4.1	AuthModule	Test:TC-Auth-01	N	
RRS-F-003	The system shall allow a logged-in user to log out, ending...	4.1	AuthModule	Test:TC-Auth-02	N	
RRS-F-004	The system shall allow a user to reset a forgotten password...	4.1	AuthModule	Test:TC-Auth-03	N	
RRS-F-010	The system shall allow a user to search for trains by source...	4.2	SearchModule	Test:TC-Search-01	N	
RRS-F-011	The system shall display the number of available seats per...	4.2	SearchModule	Test:TC-Search-02	N	
RRS-F-012	The system shall display the fare for each class when a passenger...	4.2	SearchModule	Test:TC-Fare-01	N	
RRS-F-020	The system shall allow a logged-in passenger to select a train...	4.3	BookingModule / PaymentModule	Test:TC-Book-01	N	
RRS-F-021	The system shall check seat availability before confirming...	4.3	BookingModule / PaymentModule	Test:TC-Book-02	N	
RRS-F-022	The system shall process payment through a single integrated...	4.3	BookingModule / PaymentModule	Test:TC-Pay-01	N	
RRS-F-023	The system shall generate a unique PNR and reduce the available...	4.3	BookingModule / PaymentModule	Test:TC-Pay-02	N	
RRS-F-030	The system shall allow a logged-in passenger to cancel online...	4.4	CancellationModule	Test:TC-Cancel-01	N	
RRS-F-031	The system shall increase the available seat count for the...	4.4	CancellationModule	Test:TC-Cancel-02	N	
RRS-F-032	The system shall display a cancellation confirmation message...	4.4	CancellationModule	Test:TC-Cancel-03	N	
RRS-F-040	The system shall allow a user to check the status of a booking...	4.5	PNRModule	Test:TC-PNR-01	N	
RRS-F-041	The system shall allow a logged-in passenger to view a list...	4.5	PNRModule	Test:TC-History-01	N	
RRS-F-050	The system shall allow an authorised admin to add a new train...	4.6	AdminModule	Test:TC-Admin-01	N	
RRS-F-051	The system shall allow an authorised admin to update or remove...	4.6	AdminModule	Test:TC-Admin-02	N	
RRS-F-052	The system shall allow an authorised admin to view the list...	4.6	AdminModule	Test:TC-Admin-03	N	
RRS-NF-001	Train search results shall be returned within 3 seconds under...	5	SearchModule	Test:TC-Perf-01	N	
RRS-NF-002	The user interface shall be simple and consistent, allowing...	5	WebUI	Test:TC-UX-01	N	
RRS-NF-	The system shall not allow two	5	BookingModule / DB	Test:TC-Integrity-	N	

Req ID	Requirement (short)	Section ref	Module	Test case(s)	Status (N/P/A)	Comments
003	bookings to be confirmed f...			01		
RRS-NF-004	The system shall remain available and functional for the ...	5	Infra	Test:Manual check	N	
RRS-NF-005	The codebase shall be organised into clearly separated mo...	5	All modules	Test:Manual review	N	
RRS-SR-001	User passwords shall be stored in hashed form (e.g., bcrypt...	5.1.2	AuthModule	Security test plan	N	
RRS-SR-002	The system shall use token-based session authentication (...)	5.1.2	AuthModule	Security test plan	N	
RRS-SR-003	The system shall validate and sanitise all user input on ...	5.1.2	BookingModule / AdminModule	Security test plan	N	
RRS-SR-004	Admin-only actions (add/update/delete train, view all boo...	5.1.2	AdminModule	Security test plan	N	
RRS-SR-005	All communication between the browser and the server shal...	5.1.2	Infra	Security test plan	N	